Fall 2025 EE582 Homework 02 Report
Nodal-Analysis-based Modeling of Electric Circuits

Aryan Ritwajeet Jha

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Section A (Numerical Oscillations of Discretization Rules)

One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit on the right considers a case where a dc voltage is suddenly applied to a capacitor. Because the source voltage is applied like a step function, we know that the current of the capacitor should look like an impulse, analytically. Let us assume that in the simulations all the variables are zero at $t = 0$ and the voltage source takes its value at $t = \Delta t$.

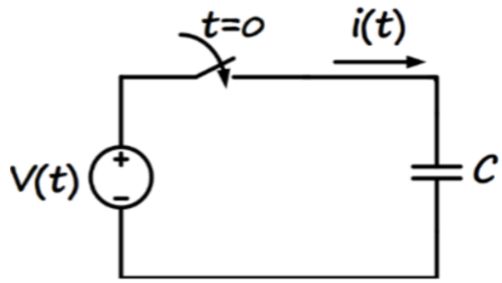

Figure 1: Simple capacitor circuit with DC voltage source

1) Determine the step-by-step solution for the current in the capacitor (in terms of capacitance C, time-step size Δt, and source voltage) using the following discretization rules: a) Forward Euler, b) Backward Euler, c) Trapezoidal.

Section A: Question 2

Comment on how the simulation time-step size impacts the numerical solution, and which method is preferred in this case.

Section B: Question 1

The parameters of the circuit below are: $R = 10\Omega, L = 20mH, e(t) = 10Vdc$, and the switch is closed at $t = 0$. The circuit is to be solved from $t = 0$ to $t = 10ms$ using:

a) PSCAD software,

b) your own MATLAB code based on the nodal analysis method.